VidMind AI

Product Feature Specification

AI-Powered YouTube Learning Companion

Project	VidMind AI — YouTube Learning Companion
Team	AgentForge (Team of 2)
Hackathon	OceanLab X CHARUSAT Hacks 2026 — 48 Hour Build
Stack	FastAPI · LangChain v1.2.4 · OpenAI GPT-3.5 · FAISS · LangSmith · React + Tailwind
Version	v2.0 — April 2026

1. Project Overview

VidMind AI transforms any YouTube video into a complete, interactive learning experience. By combining Retrieval-Augmented Generation (RAG) with intelligent tutoring behaviors, VidMind goes far beyond a simple chatbot — it acts as a personalized AI tutor that adapts to how each student thinks, asks, and learns.

The problem it solves:

- 47 million students watch YouTube lectures daily but retain very little from passive watching
- No existing tool turns a video into an active, queryable, exam-ready knowledge base
- Students waste time re-watching videos instead of targeted review

VidMind's pitch:

"Paste any YouTube lecture. Chat with it, quiz yourself, get exam-ready — all powered by AI that actually knows what the video says."

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2. Core Feature Set

VidMind AI is built around five high-value features, each chosen because students will pay for it and judges will immediately understand its value.

#	Feature	Priority
F1	RAG Chat with Timestamp-Anchored Answers	MUST
F2	Hybrid RAG — Video + World Knowledge Mode	MUST
F3	Exam Mode — Personalized Study Plan Generator	MUST
F4	Confusion Detector — Adaptive Explanation Engine	HIGH
F5	Cross-Video Playlist Intelligence	HIGH

Feature 1 — RAG Chat with Timestamp-Anchored Answers

Core Value: Every AI answer is grounded in the actual video content, with clickable timestamps that jump to the exact moment in the video.

How it works:

- YouTube URL is submitted → transcript is extracted using YouTube Transcript API
- Transcript is split into semantic chunks (1000 tokens, 200 overlap) — each chunk tagged with start timestamp and video ID
- Chunks are embedded using OpenAI text-embedding-ada-002 and stored in a FAISS vector index
- Student asks a question → query is embedded → top-4 most relevant chunks are retrieved
- GPT-3.5-turbo generates an answer using ONLY the retrieved chunks as context
- Retrieved chunks carry their timestamps → answer response includes clickable timestamp chips
- Clicking a timestamp chip opens YouTube at that exact second in a new tab

Why judges care:

This is the single most impressive live demo moment. When judges see an answer appear with "▶ 3:24 | ▶ 7:15" chips that jump to those exact video moments, it immediately communicates the product's trustworthiness. Students can verify every answer.

API Endpoint:

POST	/chat → { session_id, question } → { answer, timestamps[] }
POST	/load-video → { url } → { session_id, summary, difficulty, key_topics, study_time_minutes }

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Feature 2 — Hybrid RAG (Video + World Knowledge)

Core Value: If the video doesn't cover a student's question, VidMind transparently falls back to GPT's general knowledge — so students are never left with a dead end.

Routing Logic:

- Student submits a question
- RAG retriever scores the top-4 chunks — if relevance score < threshold (0.75), flag as 'outside video scope'
- A routing decision is made: Video-Grounded mode (high relevance) or World Knowledge mode (low relevance)
- In Video-Grounded mode: answer is generated from transcript chunks only, with timestamps
- In World Knowledge mode: GPT answers from general training knowledge, response is tagged with ' Not covered in this video — general answer'
- Student always knows the source of the information — builds trust

This feature required only 2 additional hours of implementation on top of the base RAG chain — highest ROI feature in the entire build.

Feature 3 — Exam Mode

Core Value: Student types their exam topic and available hours. VidMind generates a personalized, time-boxed study plan from the video — acting as a private tutor, not a search engine.

User Flow:

- Student types: "I have an exam on self-attention in 2 hours"
- VidMind reads the full transcript and identifies exam-relevant concepts
- GPT generates a structured JSON study plan: sections → time allocation → key concepts per section → one checkpoint question per section
- Plan is rendered as a visual timeline in the UI with estimated minutes per section
- Each section has a "Quiz me on this" button that triggers targeted quiz questions
- A confidence score is shown at the end based on quiz performance

System Prompt Strategy:

```
"You are an expert tutor. The student has {hours} hours before their exam on {topic}. Based on this video transcript, create a focused study plan. Prioritize exam-relevant concepts ruthlessly. Return JSON: { plan: [{ section, duration_mins, concepts[], checkpoint_question }] }"
```

Feature 4 — Confusion Detector

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Core Value: VidMind silently monitors how a student phrases questions. When confusion is detected, it automatically simplifies the explanation and adds a real-world analogy — without the student having to ask.

Technical Implementation:

- After the RAG answer is generated, a second lightweight GPT-3.5 call runs on the student's question
- It returns a confusion score 1–5 and a brief reason
- Score ≥ 4: RAG prompt is augmented with 'Student seems confused. Explain using a simple real-world analogy first, then the technical explanation'
- Frontend shows a subtle ' Simplified' badge on these AI messages
- Confusion patterns are tracked per session — frequent confusion flags a concept as 'needs review' in the session summary

Signals the confusion detector watches for:

Signal	Example
Vague or repeated questions	*"Can you explain that again?" / "What does that mean?"*
Contradictory follow-ups	*"But you said X, so why is it Y?"*
Filler-heavy phrasing	*"I don't really get... like... how does it work?"*
Very short follow-ups after a long answer	*"ok" / "hmm" / "what"*

Feature 5 — Cross-Video Playlist Intelligence

Core Value: Load multiple videos from a course or playlist. Ask cross-video questions. VidMind answers with source attribution — telling you which video and timestamp covers each part of the answer.

Example queries it enables:

- *"Which of the 3 videos explains backpropagation most clearly?"*
- *"Give me a combined summary of all 4 lectures from week 1"*
- *"What does video 2 say about attention that video 1 didn't cover?"*
- *"Across all videos, what are the top 5 concepts I need to know for my exam?"*

Technical Architecture:

- Each video loaded into a separate FAISS index tagged with video_id and source label
- A unified retriever searches across all indexes simultaneously
- Retrieved chunks carry source metadata → answers include citations: 'According to Video 2 at 7:15...'
- Session store tracks multiple videos per session — frontend shows loaded videos as dismissible chips

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- New endpoint: POST /add-video — accepts session_id + new URL, builds and merges new index

3. Supporting Features

These features are generated on-demand from the loaded transcript and add significant demo value:

	Feature	Description
	Video Intelligence Dashboard	Auto-generates on load: 3-sentence summary, 5 key topics, difficulty level (Beginner/Intermediate/Advanced), estimated study time in minutes
	Quiz Generator	5 MCQ questions with 4 options each, correct answer, and explanation. Interactive scoring with correct/wrong highlighting.
	Smart Study Notes	Structured markdown notes with sections, bullet points, bolded key terms, and a Key Takeaways section. One-click download as .md file.
	Flashcard Generator	8 concept/answer flashcards with 3D flip animation. Front: concept or question. Back: definition or explanation.
	Chat History Export	Full conversation exported as a formatted .txt file with timestamps referenced per AI message.
	LangSmith Observability	Every chain call — RAG, quiz, notes, confusion scoring — is traced in real time. Latency, token usage, retrieved chunks all visible in LangSmith dashboard.

4. Features Evaluated & Rejected

Two additional features were proposed and evaluated. Both were rejected for the 48-hour build:

Feature	Decision	Reason
Celebrity Voice Cloning (e.g. Shah Rukh Khan)	Rejected	Legally problematic (cloning real celebrity voice without consent). Technically requires custom TTS model training or paid API with approval. Not achievable in 48 hours.
Auth System / User Accounts	⏭ Deferred	In-memory session management is sufficient for a 48-hour demo. Auth adds 6+ hours of complexity with zero demo value for judges.

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5. Tech Stack

Layer	Technology	Purpose
LLM	OpenAI GPT-3.5-turbo	All chains: RAG, summary, quiz, notes, confusion scoring
Embeddings	OpenAI text-embedding-ada-002	Semantic chunk embeddings for FAISS retrieval
Vector Store	FAISS (in-memory)	Fast similarity search across transcript chunks
AI Framework	LangChain v1.2.4	Chain orchestration, RAG pipeline, history management
Observability	LangSmith	Real-time tracing of every chain call — demo-worthy
Backend	FastAPI + Uvicorn	REST API layer, session management, all endpoints
Frontend	React + Tailwind CSS	Component-based SPA with dark theme
UI Components	shadcn/ui	Production-grade component library
Transcripts	youtube-transcript-api	Extracts timestamped transcripts from any YouTube video
Demo Tunnel	ngrok	Exposes localhost to judges without deployment

6. 48-Hour Execution Plan

Hours	Backend (Yash)	Frontend (Teammate)
0–2	Setup: env, dependencies, GitHub repo, API keys	Setup: React + Tailwind + shadcn/ui
2–10	youtube_chatbot.py v2: timestamp RAG, summary, quiz, notes chains	Landing page, dashboard layout, chat UI shell
10–14	FastAPI endpoints: /load-video, /chat, /generate-*	Connect frontend to backend APIs, loading states
14–18	Exam Mode chain + endpoint	Exam Mode UI: topic input, timeline

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		renderer
18–22	Confusion Detector + Cross-Video Mode	Confusion badge, multi-video chips, quiz/notes UI
22–30	Integration sprint — both fix bugs together	Integration sprint — both fix bugs together
30–36	Error handling, LangSmith verification, ngrok setup	Animations, polish, mobile responsiveness
36–42	End-to-end demo rehearsal — practice the script	End-to-end demo rehearsal — time the demo
42–48	Pitch deck (both together)	GitHub README, screenshots for slides

The Iron Rule: No new features after Hour 22. A demo that works perfectly with 3 features beats a buggy demo with 8 features every single time.

7. Live Demo Script (4 minutes)

Practice this script until both team members can run it in their sleep. The demo should take no longer than 4 minutes.

Minute 0:00 – 0:30

Paste the video URL. Show the dashboard appear in ~5 seconds with auto-generated summary, difficulty badge, key topics, and study time. Say: 'The moment a video loads, VidMind reads the entire transcript and tells you what you're about to learn.'

Minute 0:30 – 1:30

Type a smart question about the video. Show the answer appear with timestamp chips. Click one chip — YouTube opens at that exact moment. Say: 'Every answer tells you exactly where in the video it came from. Students can verify anything instantly.'

Minute 1:30 – 2:15

Type 'I have an exam on this topic in 2 hours.' Show the study plan timeline render. Say: 'This is our Exam Mode — a personalized tutor that tells you what to focus on, in what order, with how much time.'

Minute 2:15 – 2:45

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Ask a vague follow-up question to trigger Confusion Detector. Show the ' Simplified' badge appear on the response. Say: 'VidMind watches how you ask questions. If you seem confused, it automatically switches to simpler explanations.'

Minute 2:45 – 3:15

Add a second video URL. Ask a cross-video question. Show source attribution in the answer. Say: 'Load an entire course. Ask questions across all lectures. VidMind knows which video said what.'

Minute 3:15 – 4:00

Switch to LangSmith dashboard on second screen. Show live traces. Say: 'This is our observability layer. Every AI call is traced — latency, token usage, retrieved context. This is production-grade.'

8. Pitch Narrative

"47 million students watch YouTube lectures daily. 90% retain almost nothing — because passive watching doesn't work. VidMind AI transforms any YouTube video into an active learning experience: contextual Q&A with source timestamps, personalized exam prep, an AI that detects when you're confused and explains it differently. It's not a chatbot. It's a tutor that lives inside every YouTube video."

Market framing:

- EdTech market: $400B globally, growing at 16% CAGR
- RAG-powered tutoring: nascent category, no dominant player
- Monetization: ₹99/month per student, ₹499/month for institutions
- Distribution: browser extension → LMS plugin → API for EdTech platforms

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